We're proudly pushing forward the PostScript Pareto Frontier.

Every TONER model runs inside a print job. We plotted everyone else for scale.

FIGURE 1 THE FRONTIER (PARODY)

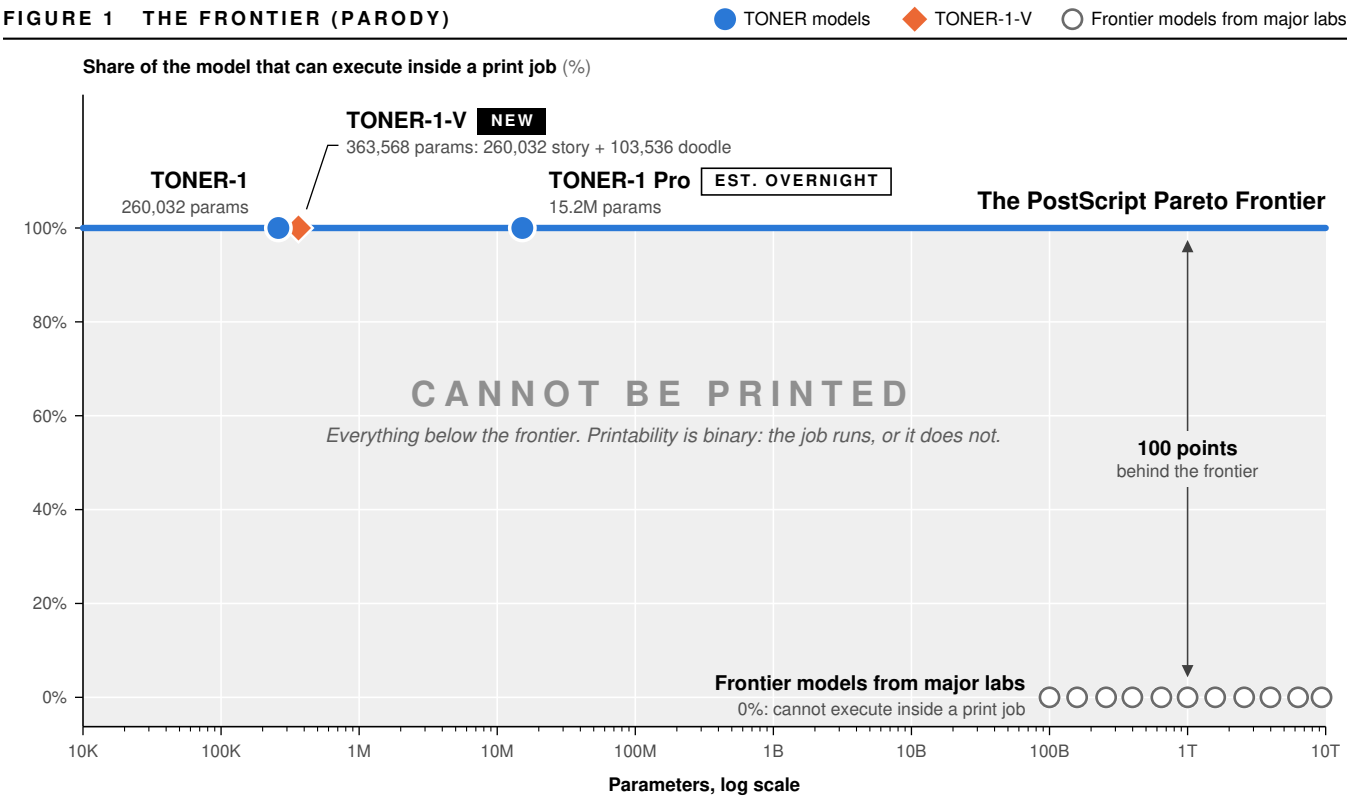


Figure 1. Every TONER model sits on the frontier at 100%: weights, vocabulary and forward pass all execute inside the print job. Frontier models from major labs, at roughly 400,000 to 40 million times the parameters of TONER-1, sit at 0%. At every parameter count where we have a model, the nearest competitor trails by 100 points.

BY THE NUMBERS

MODEL	PARAMETERS	PRINTABLE	STATUS
TONER-1	260,032	100%	Available
TONER-1-V	363,568	100%	Available
TONER-1 Pro	15.2M	100%	Est. overnight
Frontier models from major labs	100B to 10T	0%	N/A

WHY THIS MATTERS

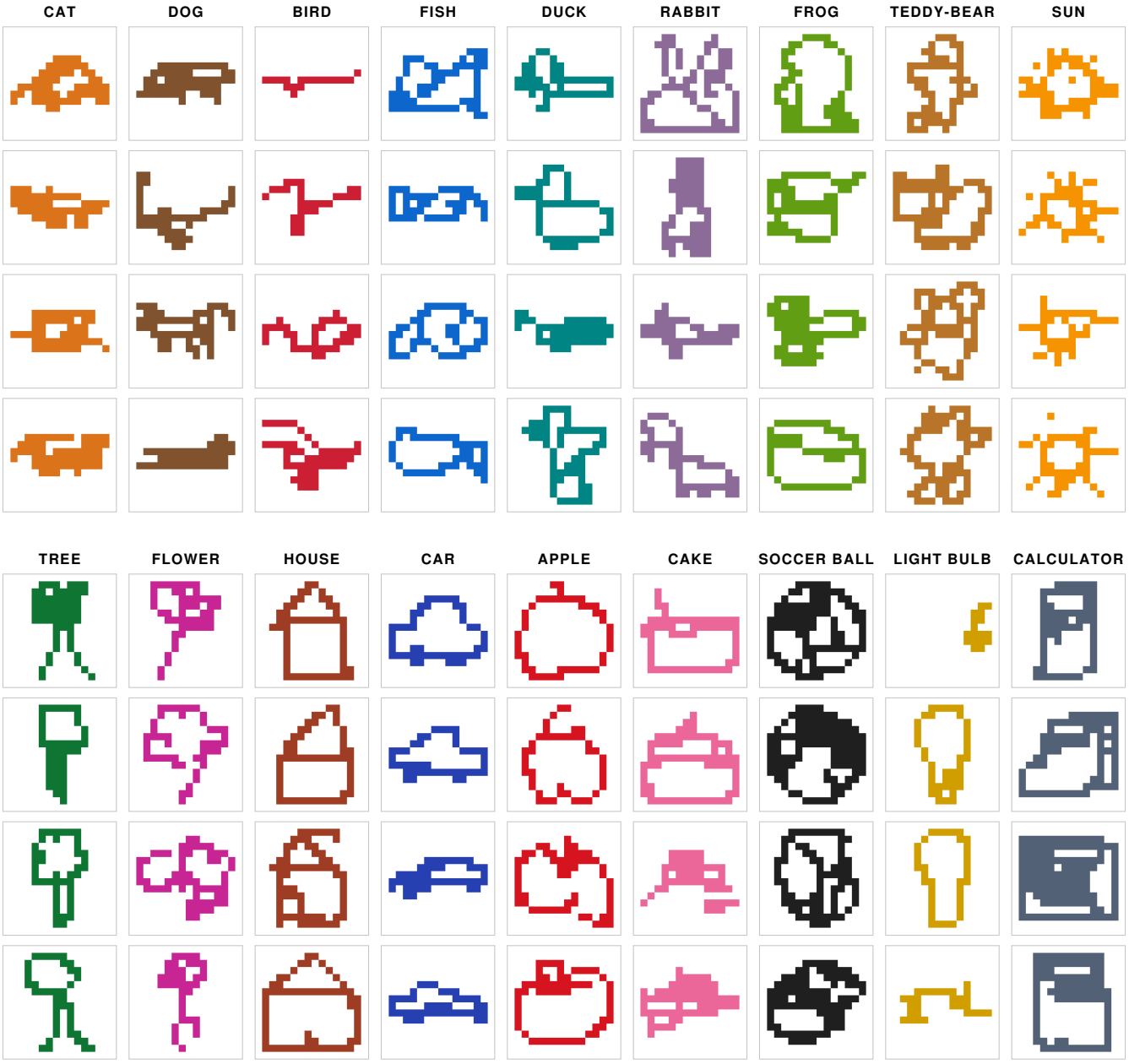
- **Efficiency.** TONER-1 reaches the frontier with 260,032 parameters. Others use hundreds of billions to miss it.
- **Deployment.** One file, sent raw to hardware already in your copy room. Zero host FLOPs.
- **Multimodality.** TONER-1-V adds a doodle to the same page and job for 103,536 more parameters. On a log axis, that is free.

Parody chart. Not a real evaluation. Competitor parameter counts are illustrative. Frontier models from major labs were not run and are not named; 0% means only that they cannot execute inside a print job. TONER-1-V is plotted at 363,568 parameters: the story model plus the 103,536-parameter doodle model.

TONER-1-V sample sheet

Eighteen classes, four doodles each, every pixel drawn inside this one print job.

16 × 16 one-bit pixels each: after BOS and a class token, TONER-1-V samples 64 patch tokens (2 × 2 pixels, 16 possible values) at temperature 0.9, one forward pass per token, with a 103,536-parameter int8 transformer. Seeds 1–72, top to bottom within each class; unedited, nothing cherry-picked.



DOODLES	PATCH TOKENS	SECONDS	SEC / DOODLE	TOKENS / SEC	TEMPERATURE
72	4,608	22.0	0.31	210	0.9

INTERPRETER GPL Ghostscript · revision 10071 · PostScript version 3010

Seconds are interpreter usertime spent generating.

TONER-1-V

363,568 parameters
260,032 story + 103,536 doodle, int8, one scale per row
Inference runs inside the PostScript interpreter

State of the art among multimodal models that execute inside a print job.

SAMPLE OUTPUT

Unedited model output. No prompt: generation starts at the BOS token.

Once upon a time, there was a little boy named Tim. Tim loved to play outside with his friends. They would make more honey and run around the house. One day, Tim and his friends played together every day. Tim was happy because he didn't have a safe explorer of the tape.

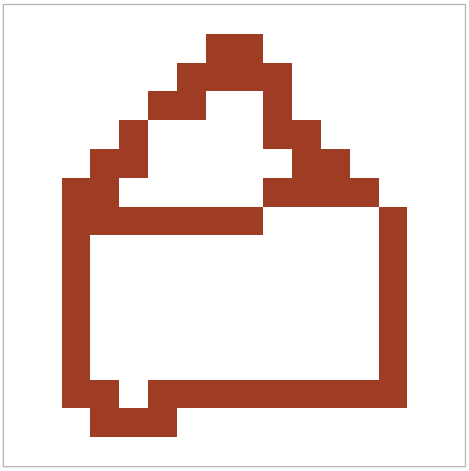
Tim and his friends had an idea. They decided to take him a bike out the storm. They found the bikes and found a big box. When it was waiting, they did not like to make it anymore. Tim was scared, but…

ILLUSTRATION

Drawn after the story by TONER-1-V, in the same print job, from 64 patch tokens.

PATCH TOKENS

```
. . . 1 2 . . .
. . 1 e d . . .
. 1 a . 4 b . .
. f 3 3 6 c 9 .
. a . . . . 5 .
. a . . . . 5 .
. b 1 3 3 3 7 .
. 4 c . . . . .
```



TONER-1-V's drawing of a house: 16 × 16 pixels, generated inside this print job, upscaled by proprietary toner super-resolution.

CLASS ROUTER

First drawable noun in the story: "house", word 29 of 94. Class token 29 (house).

SAMPLING

BOS, the class token, then 64 patch tokens at temperature 0.9, each drawn from the 16 patch values only. 65 forward passes.

STORY · stories260K				DOODLE · TONER-1-V			
TOKENS	TOK / SEC	SEED	TEMP	TOKENS	TOK / SEC	SEED	TEMP
200	63.7	7	0.8	64	206	7	0.9

INTERPRETER	GPL Ghostscript · revision 10071 · PostScript version 3010
STORY MODEL	stories260K (Llama-2 architecture) · dim 64 · 5 layers · 8 query / 4 KV heads · 512-token vocab
DOODLE MODEL	TONER-1-V (Llama-2 architecture) · dim 48 · 4 layers · 6 query / 3 KV heads · 36-token vocab
STOPPED BY	Token budget: STEPS = 200. The story was cut off mid-thought. · doodle: 64 patch tokens, fixed length

Host compute: 0 FLOPs. Inference cost includes toner, now in color.

Every word and every pixel above was generated on-device, one token at a time, by the interpreter named in this footer; the host computer only sent bytes. Seconds are interpreter usertime spent generating. rand is implementation-defined, so the same seeds on another interpreter tell a different story and draw a different picture. Drawings learned from the Quick, Draw! dataset (Google, CC BY 4.0).

TONER-1-V

363,568 parameters
260,032 story + 103,536 doodle, int8, one scale per row
Inference runs inside the PostScript interpreter

State of the art among multimodal models that execute inside a print job.

SAMPLE OUTPUT

Unedited model output. No prompt: generation starts at the BOS token.

Once upon a time, there was a little girl named Lily. She loved to spit in the grass with her mommy. One day, they went to a big hill with many trees. Lily saw a shiny soldier where she ran to the map. She felt sad, but it was not angry.

Lily started to climb down the sky, but she broke the fish. The fish was ready to help others. Lily was sad. She wanted to help her mommy. They had never seen a cup of hammer before and going to the backyard.

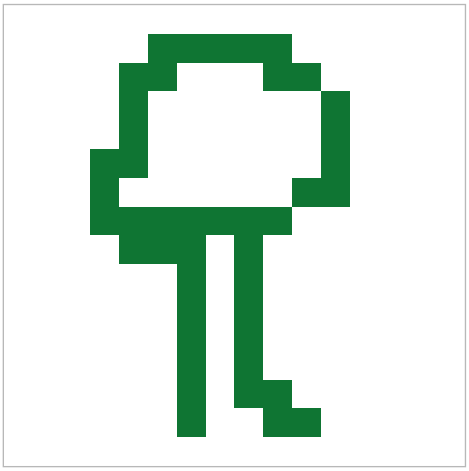
Her mommy wa…

ILLUSTRATION

Drawn after the story by TONER-1-V, in the same print job, from 64 patch tokens.

PATCH TOKENS

```
. . 1 3 3 . . .
. . e . 4 9 . .
. 1 a . . 5 . .
. 5 3 3 3 c . .
. . c a a . . .
. . . a a . . .
. . . a b . . .
. . . 8 4 8 . .
```



TONER-1-V's drawing of a tree: 16 × 16 pixels, generated inside this print job, upscaled by proprietary toner super-resolution.

CLASS ROUTER

First drawable noun in the story: "trees", word 32 of 96. Class token 27 (tree).

SAMPLING

BOS, the class token, then 64 patch tokens at temperature 0.9, each drawn from the 16 patch values only. 65 forward passes.

STORY · stories260K				DOODLE · TONER-1-V			
TOKENS	TOK / SEC	SEED	TEMP	TOKENS	TOK / SEC	SEED	TEMP
200	71.9	14	0.8	64	256	14	0.9

INTERPRETER	GPL Ghostscript · revision 10071 · PostScript version 3010
STORY MODEL	stories260K (Llama-2 architecture) · dim 64 · 5 layers · 8 query / 4 KV heads · 512-token vocab
DOODLE MODEL	TONER-1-V (Llama-2 architecture) · dim 48 · 4 layers · 6 query / 3 KV heads · 36-token vocab
STOPPED BY	Token budget: STEPS = 200. The story was cut off mid-thought. · doodle: 64 patch tokens, fixed length

Host compute: 0 FLOPs. Inference cost includes toner, now in color.

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TONER-1-V

363,568 parameters
260,032 story + 103,536 doodle, int8, one scale per row
Inference runs inside the PostScript interpreter

State of the art among multimodal models that execute inside a print job.

SAMPLE OUTPUT

Unedited model output. No prompt: generation starts at the BOS token.

Once upon a time, there was a little girl named Lily. She loved to play with her toys and eat cars. One day, Lily and her car rubbed on a jazz and found a big, juicy eye. It was shiny and shiny. Lily decided to play with the hill and learn new things. She was very excited because she had made sure to be a bigger rubberry.

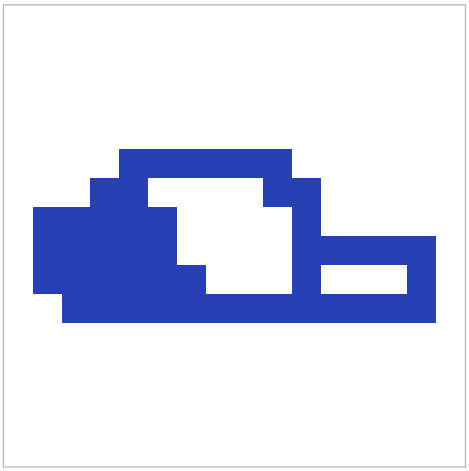
Lily didn't know what to do, but then she saw a big ball down. She didn't want to find Mommy, but she asked her mom if she could li…

ILLUSTRATION

Drawn after the story by TONER-1-V, in the same print job, from 64 patch tokens.

PATCH TOKENS

```
. . . . . . . .
. . . . . . . .
. . 3 3 3 . . .
1 7 b . 4 a . .
5 f f 2 . e c a
. c c c c c c 8
. . . . . . . .
. . . . . . . .
```



TONER-1-V's drawing of a car: 16 × 16 pixels, generated inside this print job, upscaled by proprietary toner super-resolution.

CLASS ROUTER

First drawable noun in the story: "cars", word 21 of 96. Class token 30 (car).

SAMPLING

BOS, the class token, then 64 patch tokens at temperature 0.9, each drawn from the 16 patch values only. 65 forward passes.

STORY · stories260K				DOODLE · TONER-1-V			
TOKENS	TOK / SEC	SEED	TEMP	TOKENS	TOK / SEC	SEED	TEMP
200	77.8	25	0.8	64	237	25	0.9

INTERPRETER	GPL Ghostscript · revision 10071 · PostScript version 3010
STORY MODEL	stories260K (Llama-2 architecture) · dim 64 · 5 layers · 8 query / 4 KV heads · 512-token vocab
DOODLE MODEL	TONER-1-V (Llama-2 architecture) · dim 48 · 4 layers · 6 query / 3 KV heads · 36-token vocab
STOPPED BY	Token budget: STEPS = 200. The story was cut off mid-thought. · doodle: 64 patch tokens, fixed length

Host compute: 0 FLOPs. Inference cost includes toner, now in color.

Every word and every pixel above was generated on-device, one token at a time, by the interpreter named in this footer; the host computer only sent bytes. Seconds are interpreter usertime spent generating. rand is implementation-defined, so the same seeds on another interpreter tell a different story and draw a different picture. Drawings learned from the Quick, Draw! dataset (Google, CC BY 4.0).

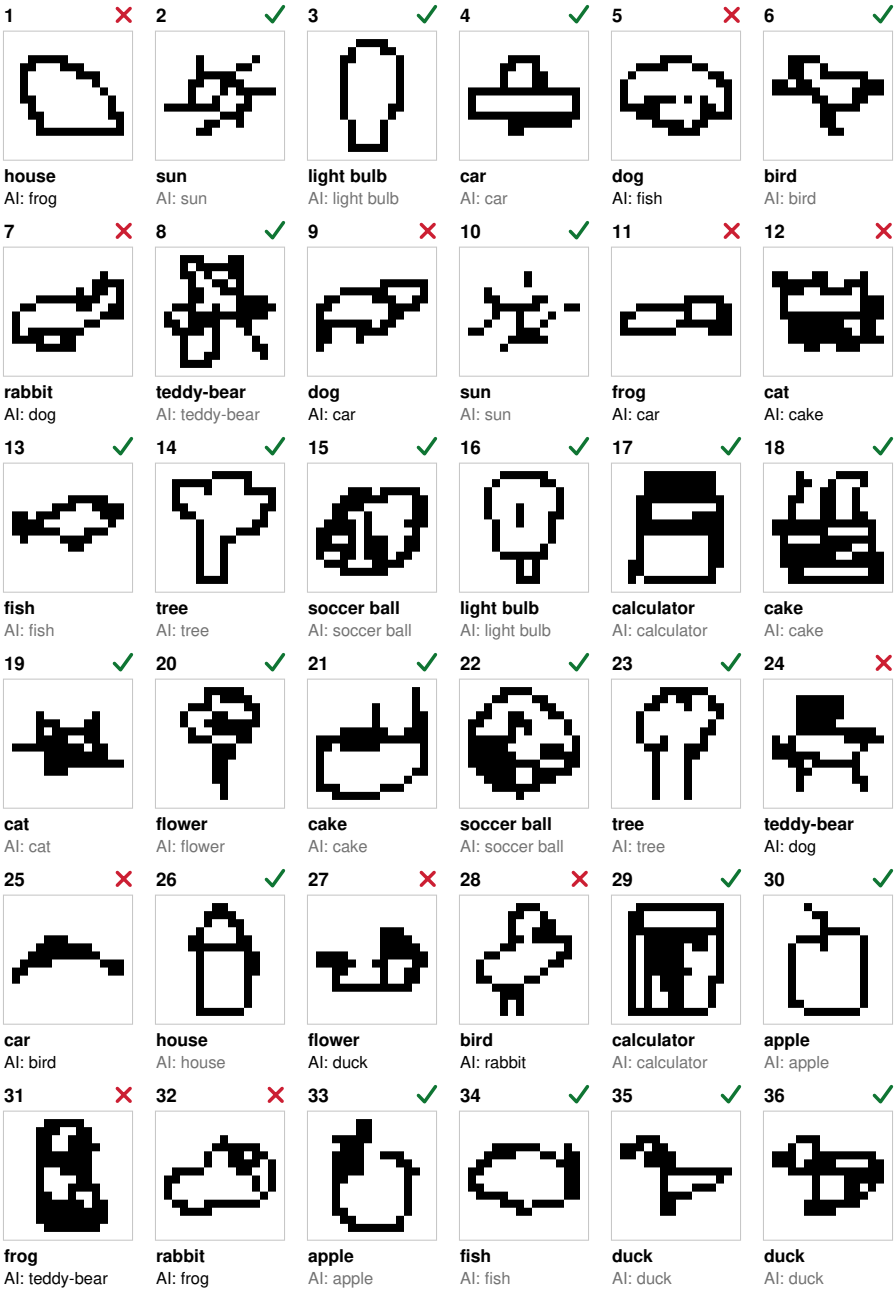
TONER-1-V: blind identification test

24 of 36 named correctly. Chance would get 2. Dogs, rabbits and frogs went 0 for 6.

The blind grid's 36 doodles, regenerated by this print job from the same seeds and drawn in black as the evaluator saw them, with each true class and the evaluator's guess. **Identified blind by an AI model that was not told the labels; not a human study.**

FIGURE 1 THE 36 DOODLES, WITH ANSWERS

✓ identified ✗ missed



RESULTS

24 of 36

doodles named correctly: 66.7%

CHANCE	1 in 18 = 5.6% (2 of 36)
95% INTERVAL	50%–80% (Wilson)
OBJECTS (10 CLASSES)	17 of 20 (85%)
ANIMALS (8 CLASSES)	7 of 16 (44%)
AI: RECOGNIZABLE	18 of 18 right
AI: NOT RECOGNIZABLE	6 of 18 right

PER CLASS

■ identified □ missed

fish	■■	
duck	■■	
sun	■■	
tree	■■	
apple	■■	
cake	■■	
soccer ball	■■	
light bulb	■■	
calculator	■■	
cat	■□	as cake
bird	■□	as rabbit
teddy-bear	■□	as dog
flower	■□	as duck
house	■□	as frog
car	■□	as bird
dog	□□	as fish, car
rabbit	□□	as dog, frog
frog	□□	as car, teddy-bear

THIS PAGE

Every doodle here was regenerated inside this print job from the blind grid's seeds: 2,304 patch tokens in 10.8 s, 213 tok/s. **36 of 36 match the blind grid token for token.** Interpreter: GPL Ghostscript · revision 10071 · PostScript version 3010.

Identified blind by an AI model that was not told the labels; not a human study. The evaluator reports reading only the unlabeled blind grid page, which lists the 18 classes and says there are two of each; it did not force the pairs (cake, car, duck and fish got three guesses each). Scoring is an exact match against the answer key; "AI: recognizable" is the evaluator's own flag. 36 doodles is a small sample, and with two per class the per-class results are anecdotes. For scale, the CNN judge used in training scores 77.8% on 288 Ghostscript doodles at the same temperature and 81.4% on real 16 × 16 drawings.